

ECS 408/608

Assignment 2

Operation on Process

Solutions

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For code repository of this assignment: [📁Operating Systems Assignments](#)

1. Problem 1

Problem statement

Implement a C/C++ program that represents a parent-child relationship between two processes. Parent and child processes do print tasks, which are as follows:

(a) Parent:

- i. Hi! I am a parent process.
- ii. My PID is <PID>.
- iii. My Child PID is <PID>.

(b) Children:

- i. Hi! I am a child process.
- ii. My PID is <PID>.
- iii. My Parent PID is <PID>.

Answers

1.1 Code:

```
1 #include <stdio.h>
2 #include <unistd.h>
3 #include <sys/types.h>
4 #include <sys/wait.h>
5
6 int main() {
7     pid_t pid;
8     pid = fork();
9
10    printf("This assignment is done by Kunwar Arpit Singh 22185.\n");
11
12    if (pid < 0){
13        printf("Forking failed for this PID.\n");
14        return 1;
15    }
16 }
```

```

17     else if (pid ==0){
18         printf("Hi! I am a child process.\n");
19         printf("My PID is %d.\n", getpid());
20         printf("My Parent PID is %d.\n", getppid());
21     }
22
23     else{
24         printf("Hi! I am a parent process.\n");
25         printf("My PID is %d.\n", getpid());
26         printf("My Child PID is %d.\n", pid);
27         wait(NULL);
28     }
29
30     return 0;
31 }

```

1.2 Output:

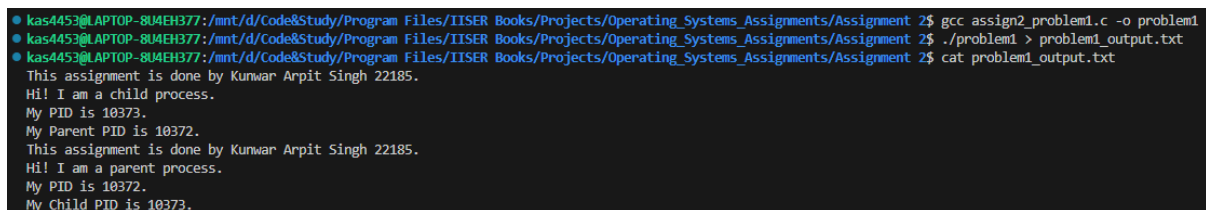
```

1 This assignment is done by Kunwar Arpit Singh 22185.
2 Hi! I am a child process.
3 My PID is 10373.
4 My Parent PID is 10372.
5 This assignment is done by Kunwar Arpit Singh 22185.
6 Hi! I am a parent process.
7 My PID is 10372.
8 My Child PID is 10373.

```

Output Screenshots

Terminal output



```

kas4453@LAPTOP-8U4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 2$ gcc assign2_problem1.c -o problem1
kas4453@LAPTOP-8U4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 2$ ./problem1 > problem1_output.txt
kas4453@LAPTOP-8U4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 2$ cat problem1_output.txt
This assignment is done by Kunwar Arpit Singh 22185.
Hi! I am a child process.
My PID is 10373.
My Parent PID is 10372.
This assignment is done by Kunwar Arpit Singh 22185.
Hi! I am a parent process.
My PID is 10372.
My Child PID is 10373.

```

2. Problem 2

Problem statement

Implement a C/C++ program with parent-child process relations that accepts n integers as input. The parent process sorts these integers using merge sort, whereas the child process sorts these integers using quick sort and prints the respective result in the format as follows:

(a) Parent:

- i. Hi! I am a parent process.
- ii. My PID is <PID>.
- iii. Merge Sort Result: <Sorted Result>.

(b) Children:

- i. Hi! I am a child process.
- ii. My PID is <PID>.
- iii. MY Parent PID is <PID>.
- iv. Quick Sort Result: <Sorted Result>.

Answers

2.1 Code:

```
1 #include <stdio.h>
2 #include <unistd.h>
3 #include <sys/types.h>
4 #include <sys/wait.h>
5
6 void printArray(int arr[], int n){
7     for (int i = 0 ; i < n ; i++){
8         printf("%d_", arr[i]);
9     }
10    printf("\n");
11 }
12
13 int partition(int arr[], int low, int high){
14     int pivot = arr[high];
15     int i = low - 1;
16
17     for (int j = low ; j < high ; j++){
18         if(arr[j] < pivot){
19             i++;
20             int temp = arr[i];
21             arr[i] = arr[j];
22             arr[j] = temp;
23     }
```

```

24     }
25
26     int temp = arr[i+1];
27     arr[i+1] = arr[high];
28     arr[high] = temp;
29
30     return i+1;
31 }
32
33 void quickSort (int arr[], int low, int high) {
34     if(low < high){
35         int pi = partition(arr, low, high);
36
37         quickSort(arr, low, pi-1);
38         quickSort(arr, pi+1, high);
39     }
40 }
41
42 void merge(int arr[], int l, int m, int r){
43     int n1 = m - l + 1;
44     int n2 = r - m;
45
46     int L[n1], R[n2];
47
48     for (int i = 0 ; i < n1 ; i++){
49         L[i] = arr[l+i];
50     }
51     for (int j = 0 ; j < n2 ; j++){
52         R[j] = arr[m+1+j];
53     }
54
55     int i = 0, j = 0, k = l;
56
57     while (i < n1 && j < n2){
58         if(L[i] <= R[j]){
59             arr[k] = L[i];
60             i++;
61         }
62         else{
63             arr[k] = R[j];
64             j++;
65         }
66         k++;
67     }
68
69     while (i < n1) {
70         arr[k] = L[i];
71         i++;
72         k++;
73     }
74

```

```

75     while (j < n2) {
76         arr[k] = R[j];
77         j++;
78         k++;
79     }
80 }
81
82 void mergeSort(int arr[], int l, int r){
83     if(l<r){
84         int m = l + (r -l) / 2;
85
86         mergeSort(arr, l, m);
87         mergeSort(arr, m+1, r);
88         merge(arr, l, m, r);
89     }
90 }
91
92 int main(){
93     printf("This assignment is done by Kunwar Arpit Singh 22185.\n");
94     pid_t pid;
95
96     int n;
97     printf("Enter number of elements: ");
98     scanf("%d", &n);
99
100    int arr[n];
101    printf("Enter %d integers.\n", n);
102    for(int i = 0 ; i < n ; i++){
103        scanf("%d", &arr[i]);
104    }
105
106    pid = fork();
107
108    if(pid < 0){
109        printf("Forking failed for this PID.\n");
110        return 1;
111    }
112    else if (pid == 0){
113        printf("Hi! I am a child process.\n");
114        printf("My PID is %d.\n", getpid());
115        printf("MY Parent PID is %d.\n", getppid());
116        quickSort(arr, 0, n-1);
117        printf("Quick Sort Result:");
118        printArray(arr, n);
119    }
120    else {
121        printf("Hi! I am a parent process.\n");
122        printf("My PID is %d.\n", getpid());
123        printf("MY Child PID is %d.\n", pid);
124        printf("Merge Sort Result:");
125        mergeSort(arr, 0, n-1);

```

```

126     printArray(arr, n);
127     wait(NULL);
128 }
129
130 return 0;
131 }

```

2.2 Output:

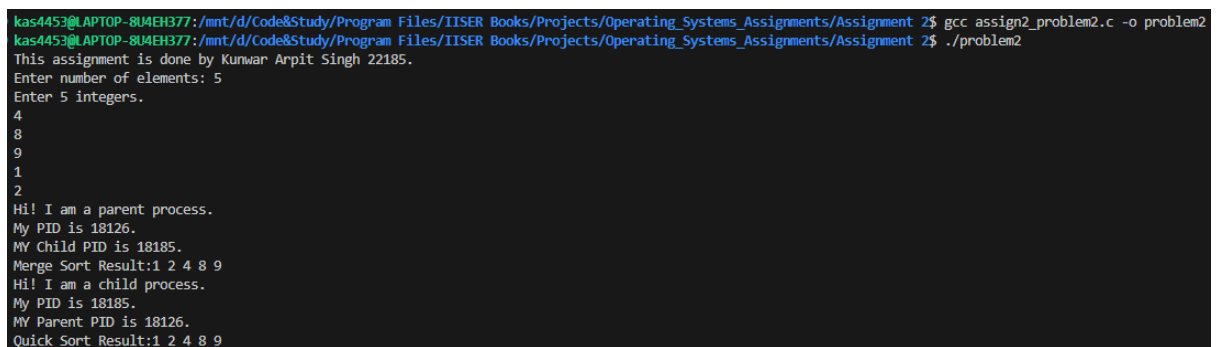
```

1 This assignment is done by Kunwar Arpit Singh 22185.
2 Enter number of elements: 5
3 Enter 5 integers.
4 4
5 8
6 9
7 1
8 2
9 Hi! I am a parent process.
10 My PID is 18126.
11 MY Child PID is 18185.
12 Merge Sort Result:1 2 4 8 9
13 Hi! I am a child process.
14 My PID is 18185.
15 MY Parent PID is 18126.
16 Quick Sort Result:1 2 4 8 9

```

Output Screenshots

Terminal output



```

kas4453@LAPTOP-8U4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 2$ gcc assign2_problem2.c -o problem2
kas4453@LAPTOP-8U4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 2$ ./problem2
This assignment is done by Kunwar Arpit Singh 22185.
Enter number of elements: 5
Enter 5 integers.
4
8
9
1
2
Hi! I am a parent process.
My PID is 18126.
MY Child PID is 18185.
Merge Sort Result:1 2 4 8 9
Hi! I am a child process.
My PID is 18185.
MY Parent PID is 18126.
Quick Sort Result:1 2 4 8 9

```

3. Problem 3

Problem statement

Write a C program with a parent-child relationship and make the child process orphan by terminating the parent process earlier than the child. The parent process performs the task of calculating the factorial of n numbers (User Input). Whereas the task performed by the child process is to generate the m (User-based input) Fibonacci numbers. Note that $m \gg n$. The output format is as follows:

(a) Parent:

- i. Hi! I am a parent process.
- ii. My PID is <PID>.
- iii. Factorial for number 5: 120.
- iv. I am terminated earlier than my child.

(b) Children:

- i. Hi! I am a child process.
- ii. My PID is <PID>.
- iii. MY Parent PID is <PID>.
- iv.

Answers

3.1 Code:

```
1 #include <stdio.h>
2 #include <unistd.h>
3 #include <stdlib.h>
4 #include <sys/types.h>
5 #include <sys/wait.h>
6
7 long long factorial(int x){
8     long long fact = 1;
9     for(int i = 1 ; i <= x ; i++){
10         fact *= i;
11     }
12     return fact;
13 }
14
15 void fibonacci(int m){
16     long long a = 0, b = 1, next;
17
18     printf("Fibonacci numbers for %d numbers: ", m);
19
20     for(int i = 0 ; i < m ; i++){
```

```

21     printf("%lld", a);
22     next = a + b;
23     a = b;
24     b = next;
25 }
26 printf("\n");
27 }
28
29 int main(){
30     printf("This assignment is done by Kunwar Arpit Singh 22185.\n");
31     pid_t pid;
32
33     int n, m;
34
35     printf("Enter the value of n (factorial count for parent).\n");
36     scanf("%d", &n);
37
38     printf("Enter the value of m (fibonacci count of child).\n");
39     scanf("%d", &m);
40
41     pid = fork();
42
43     if(pid < 0){
44         printf("Fork failed for this PID.\n");
45         return 1;
46     }
47     else if(pid == 0){
48         sleep(2);
49         printf("Hi! I am a child process.\n");
50         printf("My PID is %d.\n", getpid());
51         printf("My Parent PID is %d.\n", getppid());
52         fibonacci(m);
53     }
54     else{
55         printf("Hi! I am a parent process.\n");
56         printf("My PID is %d.\n", getpid());
57         printf("My Child PID is %d.\n", pid);
58
59         long long result = factorial(n);
60         printf("Factorial for number %d: %lld\n", n, result);
61         printf("I am terminated earlier than my child.\n");
62         exit(0);
63     }
64     return 0;
65 }

```

3.2 Output:

```

1 This assignment is done by Kunwar Arpit Singh 22185.
2 Enter the value of n (factorial count for parent).

```



```

3 5
4 Enter the value of m (fibonacci count of child).
5 10
6 Hi! I am a parent process.
7 My PID is 22249.
8 My Child PID is 22260.
9 Factorial for number 5: 120
10 I am terminated earlier than my child.
11 Hi! I am a child process.
12 My PID is 22260.
13 My Parent PID is 435.
14 Fibonacci numbers for 10 numbers: 0 1 1 2 3 5 8 13 21 34

```

Output Screenshots

Terminal output

```

kas4453@LAPTOP-8U4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 2$ gcc assign2_problem3.c -o problem3
kas4453@LAPTOP-8U4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 2$ ./problem3
This assignment is done by Kunwar Arpit Singh 22185.
Enter the value of n (factorial count for parent).
5
Enter the value of m (fibonacci count of child).
10
Hi! I am a parent process.
My PID is 22249.
My Child PID is 22260.
Factorial for number 5: 120
I am terminated earlier than my child.
Hi! I am a child process.
My PID is 22260.
My Parent PID is 435.
Fibonacci numbers for 10 numbers: 0 1 1 2 3 5 8 13 21 34

```

4. Problem 4

Problem statement

Write a C program showing the parent-child relationship with tree structure as shown in Figure 2. The process prints a message to verify the parent-child relationship. The sample output format is as shown below:

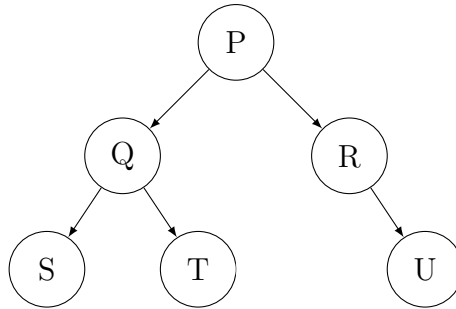


Figure 1: Parent Child Tree Relationship

(a) P:

- i. Hi! I am a root process.
- ii. My PID is <PID>.

(b) Q:

- i. Hi! I am a child process Q of Parent P.
- ii. My PID is <PID>.
- iii. MY Parent PID is <PID>.

(c) R:

- i. Hi! I am a child process R of Parent P.
- ii. My PID is <PID>.
- iii. MY Parent PID is <PID>.

(d) S:

- i. Hi! I am a child process S of Parent Q.
- ii. My PID is <PID>.
- iii. MY Parent PID is <PID>.
- iv. My Grand Parent PID is <PID>. (Optional)

(e) T:

- i. Hi! I am a child process T of Parent Q.
- ii. My PID is <PID>.
- iii. MY Parent PID is <PID>.

iv. My Grand Parent PID is <PID>. (Optional)

(f) U:

i. Hi! I am a child process U of Parent R.

ii. My PID is <PID>.

iii. MY Parent PID is <PID>.

iv. My Grand Parent PID is <PID>. (Optional)

Answers

4.1 Code:

```
1 #include <stdio.h>
2 #include <unistd.h>
3 #include <stdlib.h>
4 #include <sys/types.h>
5 #include <sys/wait.h>
6
7 int main(){
8     printf("This assignment is done by Kunwar Arpit Singh 22185.\n");
9
10    printf("Hi! I am a root process.\n\n");
11    fflush(stdout);
12    printf("My PID is %d.\n", getpid());
13    fflush(stdout);
14
15    pid_t pidQ = fork();
16
17    if(pidQ == 0){
18        printf("Hi! I am a child process of Q of Parent P.\n");
19        fflush(stdout);
20        printf("My PID is %d.\n", getpid());
21        fflush(stdout);
22        printf("My Parent PID is %d.\n", getppid());
23        fflush(stdout);
24
25        pid_t grandParentPID = getppid();
26
27        pid_t pidS = fork();
28
29        if(pidS == 0){
30            printf("Hi! I am a child process S of Parent Q.\n");
31            fflush(stdout);
32            printf("My PID is %d.\n", getpid());
33            fflush(stdout);
34            printf("My Parent PID is %d.\n", getppid());
35            fflush(stdout);
36            printf("My Grand Parent PID is %d.\n", grandParentPID);
37            fflush(stdout);
```

```

38         exit(0);
39     }
40
41     pid_t pidT = fork();
42
43     if(pidT == 0){
44         printf("Hi! I am a child process T of Parent Q.\n");
45         fflush(stdout);
46         printf("My PID is %d.\n", getpid());
47         fflush(stdout);
48         printf("My Parent PID is %d.\n", getppid());
49         fflush(stdout);
50         printf("My Grand Parent PID is %d.\n", grandParentPID);
51         fflush(stdout);
52         exit(0);
53     }
54     wait(NULL);
55     wait(NULL);
56     exit(0);
57 }
58
59 else{
60     pid_t pidR = fork();
61
62     if(pidR == 0){
63         printf("Hi! I am a child process R of Parent P.\n");
64         printf("My PID is %d.\n", getpid());
65         printf("My Parent PID is %d.\n", getppid());
66
67         pid_t grandParentPID = getppid();
68         pid_t pidU = fork();
69
70         if(pidU == 0){
71             printf("Hi! I am a child process U of Parent R.\n");
72             printf("My PID is %d.\n", getpid());
73             printf("My Parent PID is %d.\n", getppid());
74             printf("My Grand Parent PID is %d.\n", grandParentPID);
75             exit(0);
76         }
77
78         wait(NULL);
79         exit(0);
80     }
81
82 }
83 wait(NULL);
84 wait(NULL);
85 return 0;
86 }

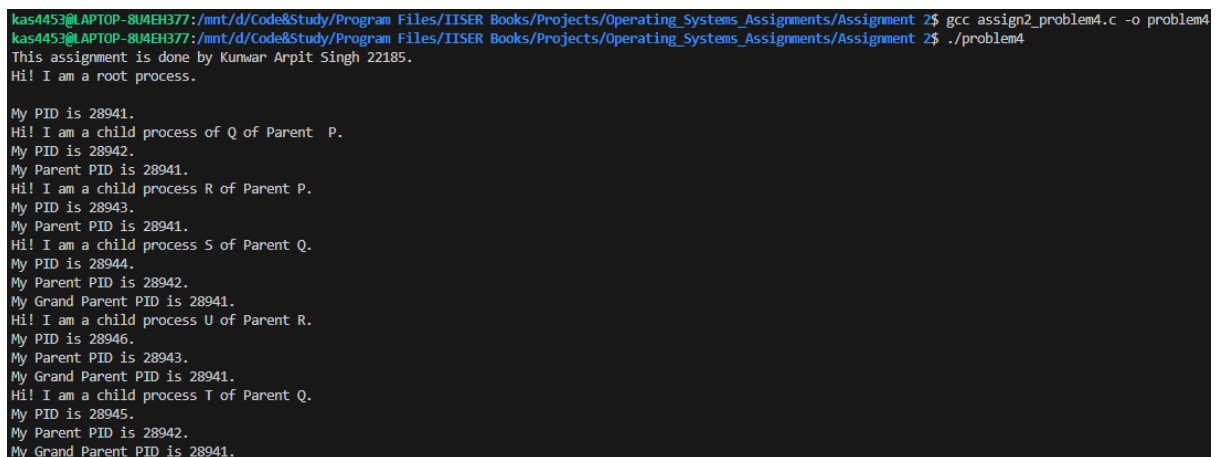
```

4.2 Output:

```
1 This assignment is done by Kunwar Arpit Singh 22185.
2 Hi! I am a root process.
3
4 My PID is 28941.
5 Hi! I am a child process of Q of Parent P.
6 My PID is 28942.
7 My Parent PID is 28941.
8 Hi! I am a child process R of Parent P.
9 My PID is 28943.
10 My Parent PID is 28941.
11 Hi! I am a child process S of Parent Q.
12 My PID is 28944.
13 My Parent PID is 28942.
14 My Grand Parent PID is 28941.
15 Hi! I am a child process U of Parent R.
16 My PID is 28946.
17 My Parent PID is 28943.
18 My Grand Parent PID is 28941.
19 Hi! I am a child process T of Parent Q.
20 My PID is 28945.
21 My Parent PID is 28942.
22 My Grand Parent PID is 28941.
```

Output Screenshots

Terminal output



```
kas4453@LAPTOP-8U4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 2$ gcc assign2_problem4.c -o problem4
kas4453@LAPTOP-8U4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 2$ ./problem4
This assignment is done by Kunwar Arpit Singh 22185.
Hi! I am a root process.

My PID is 28941.
Hi! I am a child process of Q of Parent P.
My PID is 28942.
My Parent PID is 28941.
Hi! I am a child process R of Parent P.
My PID is 28943.
My Parent PID is 28941.
Hi! I am a child process S of Parent Q.
My PID is 28944.
My Parent PID is 28942.
My Grand Parent PID is 28941.
Hi! I am a child process U of Parent R.
My PID is 28946.
My Parent PID is 28943.
My Grand Parent PID is 28941.
Hi! I am a child process T of Parent Q.
My PID is 28945.
My Parent PID is 28942.
My Grand Parent PID is 28941.
```